01 Setup - Environment Setup and Data Verification

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0.1 Introduction

This notebook checks that all required packages, data directories, and file schemas are in place before the pipeline runs.

0.2 Required Packages

All required packages are checked and missing ones are installed automatically.

0.3 Data Directories

MIMIC-IV (Johnson et al., 2023) and eICU-CRD (Pollard et al., 2018) must be accessible at the paths defined in `config.R`.

0.4 Key File Spot-Check

Checks that key source tables exist and reports their file sizes.

0.5 Schema Verification

Confirms that key columns exist and that `anchor_year_group` (needed for temporal splitting) is present.

```
## icustays columns:
## [1] "subject_id"     "hadm_id"        "stay_id"        "first_careunit"
## [5] "last_careunit"  "intime"         "outtime"        "los"
## anchor_year_group present: TRUE
```

Table 1: Package availability

	package	status
data.table	data.table	OK
arrow	arrow	OK
dplyr	dplyr	OK
tidyr	tidyr	OK
ggplot2	ggplot2	OK
lubridate	lubridate	OK
survival	survival	OK
nnet	nnet	OK
VGAM	VGAM	OK
discSurv	discSurv	OK
rms	rms	OK
cmprsk	cmprsk	OK
xgboost	xgboost	OK
pmsampsize	pmsampsize	OK
dcurves	dcurves	OK
timeROC	timeROC	OK
CalibrationCurves	CalibrationCurves	OK
purrr	purrr	OK
tibble	tibble	OK
readr	readr	OK

Table 2: Data directory status

directory	path	status
MIMIC-IV root	/Users/arif/Desktop/RESEARCH/data/mimic-iv-3.1	TRUE
MIMIC-IV/hosp	/Users/arif/Desktop/RESEARCH/data/mimic-iv-3.1/hosp	TRUE
MIMIC-IV/icu	/Users/arif/Desktop/RESEARCH/data/mimic-iv-3.1/icu	TRUE
eICU root	/Users/arif/Desktop/RESEARCH/data/eicu-collaborative-research-database-2.0	TRUE

Table 3: MIMIC-IV key files

table	exists	size_mb
admissions	TRUE	19.9
patients	TRUE	2.8
icustays	TRUE	3.3
chartevents	TRUE	3502.6
labevents	TRUE	2593.0
prescriptions	TRUE	606.7
microbiologyevents	TRUE	117.2
inputevents	TRUE	401.3

Table 4: eICU key files

table	exists	size_mb
patient	TRUE	10.4
lab	TRUE	516.8
medication	TRUE	134.3
vitalPeriodic	TRUE	1784.2
microLab	TRUE	0.1
diagnosis	TRUE	35.2

Table 5: Setup summary

item	value
Project root	/Users/arif/Desktop/RESEARCH
MIMIC-IV	AVAILABLE
eICU	AVAILABLE
Output dir	/Users/arif/Desktop/RESEARCH/sepsis/project/output
R version	4.5.2
Label variants	A (Narrow), B (Seymour-Standard), C (Liberal)

0.6 Output Directory Writability

Verifies the output directory is writable.

```
## Output dir writable: OK
## /Users/arif/Desktop/RESEARCH/sepsis/project/output
```

0.7 Summary

Everything the later stages depend on, collected into one table: where the project root resolved to, which of the two databases were found, and the R version this run will be recorded against.

```
##
## Saved: 01_setup_file_report.parquet
```